Model MEML 75:

A phenomenological parametrization of the Standard Model fermion sector from $T^2/\mathbb{Z}_3$ geometry with axion dark matter

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Abstract

We present model MEML 75, which describes the Standard Model fermion sector through the geometry of the six-dimensional orbifold $T^2/\mathbb{Z}_3$ with two independent tori. The model fits **9 quark observables** (5 masses + 3 CKM + J) with **11 free parameters** ($N_{\rm dof} = 9 - 11 = -2$ in the quark sector).

The main testable prediction: axion-like dark matter with $f_a = 1.9801 \times 10^{11}$ GeV, $m_a = 28.407$ μeV, $\nu_a = 6.8688$ GHz, and axion-photon coupling $g_{a\gamma\gamma} = +4.38 \times 10^{-15}$ GeV^{-1} (DFSZ) or -1.13×10^{-14} GeV^{-1} (KSVZ), testable at ADMX/HAYSTAC (0.5–10 GHz). The relic density $\Omega_a h^2$ is reproduced with one free parameter θ_i.

Classification of quantities: $\alpha_G = 2/(3\pi)$ is a *postulate* with geometric motivation; $\varepsilon = (\alpha_G/(4\pi))^2$ is a one-loop *ansatz*; m_{KK}, M_6 are conditionally derived at $\mathcal{I} = 0.3653$.

Fermion sector. A global fit with loss function $\chi^2 = \sum_i ((O_i - O_i^{\rm target})/\sigma_i)^2$ yields $\chi^2 = 6.195 \times 10^{-20}$ at $\tau_u = -0.1577944085 + 2.3851782590\,i$, $\tau_d = +0.3693905617 + 1.5410496057\,i$. Independent reproduction is given in §8 and Appendix C. $\mathrm{Im}\,\tau$ is stable ($\pm 0.3\%$), $\mathrm{Re}\,\tau$ is soft-flat (Appendix E).

Systematic tests (all negative): (1) the universal Yukawa ansatz is excluded; (2) 36 matrix structures are equally unstable (spread of $|V_{ub}|$ from -97% to $+25407\%$); (3) θ_i from inflation is excluded at 26σ; (4) modular forms at $\tau = e^{i\pi/3}$ do not yield α_G exactly ($|Y_2^{(6)}|^2 = 0.445302 \neq 4/9$).

Coefficient degeneracy. At $N_{\rm dof} = 9 - 11 = -2$ the coefficients c_i are non-unique: χ^2 is invariant under $c_i \to k c_i$ (Appendix D). The model *interpolates* quark observations, it does not predict them.

Verdict: the model is *phenomenological*.

1 Introduction

1.1 Motivation

Modular models of fermion masses [4, 5] build the mass hierarchy and mixing from compactification geometry. In model MEML 75, quarks and leptons are localized on two independent tori $T^2/\mathbb{Z}_3$ with moduli τ_u and τ_d, and the mass matrices are built from A_4 modular forms of weights $(2, 4, 6)$. The axion arising from the same geometric ingredients provides dark matter.

1.2 Self-containment

The paper is constructed so that the reader can reproduce *every* number without external sources:

- all Y-forms (q-expansion coefficients) — in Appendix A;
- all PDG target values — in §8;
- DE-optimization procedure with parameters — in Appendix B;
- three independent fits — in Appendix C;
- proof of scale-degeneracy — in Appendix D;
- flatness procedure — in Appendix E;
- LOO procedure — in Appendix F;
- systematic search for α_G (392 combinations) — in Appendix G.

1.3 Conventions

Table 1: Status of quantities.

Status	Meaning
[POSTULATE]	Postulate with geometric motivation
[ANSATZ]	Phenomenological ansatz
[COND]	Conditionally derived (at $\mathcal{I}$)
[MATCH]	Structural matching relation
[DERIVED]	Algebraic consequence
[FIT]	From numerical fit
[OBSERVED]	Experimental value

2 Geometry of $T^2/\mathbb{Z}_3$

2.1 Metric and modular parameter

The orbifold $T^2/\mathbb{Z}_3$:

$$z = x + \tau y, \quad ds^2 = dx^2 + dx\,dy + dy^2, \tag{2.1}$$

where $\tau = e^{i\pi/3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}$.

2.2 Fixed points

Z_3-action $z \to \omega z$, $\omega = e^{2\pi i/3}$:

$$z_1 = (0,0), \quad z_2 = (\tfrac{1}{3}, \tfrac{1}{3}), \quad z_3 = (\tfrac{2}{3}, \tfrac{2}{3}). \tag{2.2}$$

These three points provide a topological explanation for three generations.

2.3 Two-torus structure

Quarks and leptons are localized on different tori with moduli:

$$\boxed{\tau_u = -0.1577944085 + 2.3851782590\,i, \quad \tau_d = +0.3693905617 + 1.5410496057\,i} \tag{2.3}$$

Status: [FIT]. Determined by the global fit (§8, Appendix B). The values are given to 10 significant digits because the minimum is razor-thin: rounding to 3 significant digits raises χ^2 from 6×10^{-20} to 2.6×10^{-3} (Appendix C). $\mathrm{Im}\,\tau_u = 2.385$, $\mathrm{Im}\,\tau_d = 1.541$ are stable across all fits; $\mathrm{Re}\,\tau$ is soft-flat (Appendix E).

3 Bulk coupling α_G (postulate)

3.1 Definition

$$\boxed{\alpha_G = \frac{2}{3\pi} = 0.2122065908} \tag{3.1}$$

Status: [POSTULATE]. Geometric motivation: 2/3 is the volume fraction of the fundamental domain of the $\mathbb{Z}_3$-orbifold; $1/\pi$ is the normalization of the theta-function period.

3.2 Algebraic identity

$$\frac{2\pi}{3\alpha_G} = \pi^2 = 9.8696044011. \tag{3.2}$$

This is an identity, not a coincidence. Numerically: $e^{\pi^2} = 1.933369 \times 10^4$.

3.3 Check of modular forms at $\tau = e^{i\pi/3}$

At $\tau = e^{i\pi/3}$ (60 digits, 30 terms, Appendix A):

Table 2: A_4 modular forms at $\tau = e^{i\pi/3}$.

Form	Modulus	Status
$\lvert Y_1^{(2)}\rvert$	0.948674	—
$\lvert Y_2^{(2)}\rvert$	0.974090	—
$\lvert Y_3^{(2)}\rvert$	0.476383	—
$\lvert Y_1^{(4)}\rvert$	1.364023	—
$\lvert Y_2^{(4)}\rvert$	0.697153	—
$\lvert Y_3^{(4)}\rvert$	1.400784	—
$\lvert Y_1^{(6)}\rvert$	0.297141	—
$\lvert Y_2^{(6)}\rvert$	0.667310	—
$\lvert Y_3^{(6)}\rvert$	0.679090	—

Checks of α_G:

- $|Y_2^{(6)}|/\pi = 0.212411$ (deviation $+0.0965\%$);
- $|Y_1^{(6)}|/|Y_3^{(4)}| = 0.212125$ (deviation -0.0385%).

The exact equality $|Y_2^{(6)}| = 2/3$ *does not hold*: $|Y_2^{(6)}|^2 = 0.445302 \neq 4/9 = 0.444444$, $|\Delta| = 8.576 \times 10^{-4}$, stable at $n = 5, 11, 20, 30$.

3.4 Systematic search for a derivation of α_G

We performed an exhaustive search over the basis $\mathcal{B} = \{Y_i^{(2)}, Y_i^{(4)}, Y_i^{(6)}, \eta, \pi, \sqrt{\pi}, \pi^2, e^{-\pi^2}\}_{i=1,2,3}$ at $\tau = e^{i\pi/3}$. Here $\eta(\tau) = q^{1/24}\prod_{n=1}^{\infty}(1-q^n)$ is the Dedekind eta function, $q = e^{2\pi i\tau}$; $|\eta(e^{i\pi/3})| = 0.800579$.

Tested $N = 392$ combinations. Precision: 30 terms, 60-digit arithmetic.

Found 2 coincidences below 0.1%:

Table 3: Structural near-coincidences.

Combination	Value	Deviation	Stability
$\lvert Y_1^{(6)}\rvert/\lvert Y_3^{(4)}\rvert$	0.212125	$\mathbf{-0.0385}\%$	$n = 5..30$
$\lvert Y_2^{(6)}\rvert/\pi$	0.212411	$\mathbf{+0.0965}\%$	$n = 5..30$

Statistical estimate. The expected number of hits at $N = 392$ with tolerance $\pm 0.1\%$: $\langle n\rangle = 392 \cdot 0.002 = 0.78$. Observed $n_{\text{obs}} = 2$. Poisson probability:

$$P(n \geq 2 \mid \langle n\rangle = 0.78) = \frac{\gamma(2, 0.78)}{\Gamma(2)} = 0.185. \tag{3.3}$$

This is **not significant** ($p = 0.185 > 0.05$).

Conclusion. $\alpha_G = 2/(3\pi)$ remains a **postulate**.

4 Suppression factor ε (ansatz)

$$\boxed{\varepsilon = \left(\frac{\alpha_G}{4\pi}\right)^2 = \frac{1}{36\pi^4} = 2.851662 \times 10^{-4}} \tag{4.1}$$

Status: [ANSATZ]. Motivation: one-loop structure of QFT.

4.1 Matching 6D↔4D

Holomorphic Yukawa coupling: $Y_{\text{hol}}(\tau) = \sum_n N_n e^{-2\pi n \operatorname{Im}\tau} P_n(\tau)$. After canonical normalization:

$$Y_{\text{phys}} \sim e^{-S_{\text{inst}}}\,(\operatorname{Im}\tau)^{\Delta k}, \quad \Delta k = \tfrac{k_i+k_j+k_k}{2} - b, \tag{4.2}$$

with $\Delta k = 1$ (up), $\Delta k = 2$ (down).

Numerical check: $e^{-\pi^2} = 5.1723 \times 10^{-5}$; $\varepsilon/e^{-\pi^2} = 5.514$; $(2\operatorname{Im}\tau_u)^1 = 4.774$; $(2\operatorname{Im}\tau_d)^2 = 6.248$. $5.514 \in [4.774, 6.248]$ — within range.

5 Conditionally derived quantities

$$f_a = M_{Pl}^{\rm red}\,\varepsilon^2 = 1.9801 \times 10^{11}\ \text{GeV}, \tag{5.1}$$

$$m_a = \frac{(\chi^{1/4})^2}{f_a} = \frac{(75.0\ \text{MeV})^2}{f_a} = 28.407\ \mu\text{eV}, \tag{5.2}$$

$$\nu_a = \frac{m_a c^2}{h} = 6.8688\ \text{GHz}. \tag{5.3}$$

Here $\chi^{1/4} \approx 75.0$ MeV is the fourth root of the topological susceptibility of QCD, which is the correct quantity entering the axion mass formula (not the QCD scale $\Lambda_{\rm QCD} \approx 24$ MeV).[1]

6 Axion sector

6.1 Axion parameters

$$f_a = M_{Pl}^{\rm red}\,\varepsilon^2 = 1.9801 \times 10^{11}\ \text{GeV}, \tag{6.1}$$

$$m_a = \frac{(\chi^{1/4})^2}{f_a} = \frac{(75.0\ \text{MeV})^2}{f_a} = 28.407\ \mu\text{eV}, \tag{6.2}$$

$$\nu_a = \frac{m_a c^2}{h} = 6.8688\ \text{GHz}. \tag{6.3}$$

Here $\chi^{1/4} \approx 75.0$ MeV is the fourth root of the topological susceptibility of QCD, which is the correct quantity entering the axion mass formula (not the QCD scale $\Lambda_{\rm QCD} \approx 24$ MeV).[2]

Status: [COND]. ν_a within ADMX/HAYSTAC range.

6.2 Axion-photon coupling

$$g_{a\gamma\gamma} = \frac{\alpha_{\rm EM}}{2\pi f_a}\left(\frac{E}{N} - 1.92\right). \tag{6.4}$$

At $f_a = 1.9801 \times 10^{11}$ GeV: $\alpha_{\rm EM}/(2\pi f_a) = 5.866 \times 10^{-15}$ GeV^{-1}.

Table 4: Prediction for $g_{a\gamma\gamma}$.

Scenario	E/N	$g_{a\gamma\gamma}$, GeV^{-1}
KSVZ	0	-1.126×10^{-14}
DFSZ	8/3	$+4.379 \times 10^{-15}$

Both values lie within the sensitivity of ADMX Run 1D–1E. The factor of 2.6 between them means that the model predicts a *two-point* target.

[1] Using $\chi^{1/4} = 75.5$ MeV would give $m_a = 28.8$ μeV and $\nu_a = 6.96$ GHz; both values lie within the theoretical uncertainty of $\chi^{1/4}$ and are testable at ADMX/HAYSTAC.

[2] Using $\chi^{1/4} = 75.5$ MeV would give $m_a = 28.8$ μeV and $\nu_a = 6.96$ GHz; both values lie within the theoretical uncertainty of $\chi^{1/4}$ and are testable at ADMX/HAYSTAC.

7 A remark on $SU(5)$

While $SU(5)$ is the unique rank-4 group admitting the required SM content, the fermion sector here is built on A_4 modular forms. The $SU(5)$ structure is not used for the mass generation mechanism and is mentioned only for completeness.

8 Fermion sector (11 parameters)

8.1 Matrix structure

$$M_u = \begin{pmatrix} a_u Y_1^{(2)} & b_u Y_3^{(4)} & c_u Y_2^{(6)} \\ a_u Y_3^{(2)} & b_u Y_2^{(4)} & c_u Y_1^{(6)} \\ a_u Y_2^{(2)} & b_u Y_1^{(4)} & c_u Y_3^{(6)} \end{pmatrix}, \; M_d = \begin{pmatrix} a_1 Y_1^{(6)} & b Y_3^{(2)} & c Y_2^{(2)} \\ a_1 Y_3^{(6)} & b Y_2^{(2)} & c Y_1^{(2)} \\ a_2 Y_2^{(2)} & b Y_1^{(2)} & c Y_3^{(4)} \end{pmatrix}. \tag{8.1}$$

8.2 Loss function and target values

$$\chi^2 = \sum_{i=1}^{9} \left(\frac{O_i(\tau_u, \tau_d, c) - O_i^{\text{target}}}{\sigma_i} \right)^2 . \tag{8.2}$$

Table 5: Target values (PDG 2024).

Observable	$O_i^{\text{target}} \pm \sigma_i$
m_u/m_t	$(1.27 \pm 0.30) \times 10^{-3}/172.0$
m_c/m_t	$(0.620 \pm 0.020)/172.0$
m_d/m_b	$(2.70 \pm 0.20) \times 10^{-3}/2.860$
m_s/m_b	$(55.0 \pm 5.0) \times 10^{-3}/2.860$
m_b/m_t	$(2.860 \pm 0.030)/172.0$
$\lvert V_{us}\rvert$	0.22501 ± 0.00068
$\lvert V_{cb}\rvert$	0.04053 ± 0.00042
$\lvert V_{ub}\rvert$	0.00382 ± 0.00020
$\lvert J\rvert$	$(3.080 \pm 0.130) \times 10^{-5}$

8.3 Results

Independent optimization (DE, 3 seeds, Appendix B):

$$\boxed{\chi^2 = 6.195 \times 10^{-20} \quad \text{at} \quad \tau_u = -0.1577944085 + 2.3851782590\,i, \; \tau_d = +0.3693905617 + 1.5410496057} \tag{8.3}$$

Coefficients:

$$c^{\text{fit}} \approx (5.31 \times 10^{-5},\ 7.19,\ 0.039,\ 2.43 \times 10^{-4},\ 0.503,\ 1.37 \times 10^{-3},\ 2.22 \times 10^{-3}). \tag{8.4}$$

Per-observable pulls at these values: all $< 10^{-9}\sigma$.

8.4 Independent reproduction

Three independent 11-parameter DE runs (Appendix C) converge to $\mathrm{Im}\,\tau_u = 2.385$, $\mathrm{Im}\,\tau_d = 1.541$ to 0.3%, with $\chi^2 < 10^{-18}$ in all runs.

8.5 Stability of CKM under shifts in τ

With coefficients (8.4) held fixed, a shift in τ_u or τ_d by $\pm 5\%$ moves the CKM elements outside 1σ: $\mathrm{Im}\,\tau_u = 2.385 \pm 0.007$, $\mathrm{Im}\,\tau_d = 1.541 \pm 0.005$; $\mathrm{Re}\,\tau_d \in [-0.37, +0.37]$ (soft flatness, Appendix E); $\mathrm{Re}\,\tau_u \in [-0.16, -0.13]$.

Consequence for flux. Only $\mathrm{Im}\,\tau$ requires precise stabilization ($\pm 3\%$).

9 Test B.1: universal ansatz

Table 6: Levels of coefficient freedom.

Level	Parametrization	k_{coef}	N_{dof}	χ^2
L1	$a_u = b_u = c_u = y_u$	2	+3	5.28×10^6
L2	$+r$	3	+2	3.51×10^2
L3	$+r_u$	4	+1	4.45×10^3
L4	up + 1 ratio down	5	0	3.53×10^3
L5	2 ratios per sector	6	-1	4.36×10^2
L6	**7 free**	**7**	$\mathbf{-2}$	$\mathbf{2.18 \times 10^{-19}}$

Conclusion. The universal ansatz is **excluded**. Only L6 yields the ideal fit.

10 Test C.3: scan of matrix structures

We tested 36 combinations (6 up $\times$ 6 down) with 6 seeds.

Table 7: LOO for U6×D6.

Seed	χ^2	$\lvert V_{ub}\rvert$ err	$\lvert J\rvert$ err
0	8.16×10^{-9}	+25406.7%	−97.2%
1	2.09×10^{-8}	+24308.0%	+20318.7%
2	1.33×10^{-8}	+33.1%	−97.5%
3	1.04×10^{-10}	+8.4%	+19.2%
4	9.41×10^{-10}	+25302.9%	+872.6%
5	1.43×10^{-9}	+25311.8%	+570.6%

Conclusion. All 36 structures are unstable. The problem lies in over-determination ($N_{\mathrm{dof}} = -2$).

11 Axion dark matter

11.1 Analytical formula

$$\Omega_a h^2 \approx 0.12\,\theta_i^2\, f_{\mathrm{anh}}(\theta_i) \left(\frac{f_a}{3 \times 10^{11}}\right)^{7/6}, \quad f_{\mathrm{anh}}(\theta_i) = [\ln(e/(1-(\theta_i/\pi)^2))]^{7/6}. \tag{11.1}$$

11.2 Full decomposition

Step 1 (bare): $\Omega_a h^2_{\rm bare} = 0.12 \cdot 1.2459 \cdot 0.6159 = 0.0921$.

Step 2 (Lyth–Turner anharmonicity): $f_{\rm anh}(1.1162) = 1.1591$, $\Omega_a h^2_{\rm anh} = 0.1067$.

Step 3 (T-effects): $\Omega_a h^2_{\rm num} = 0.1067 \cdot (1 + \delta_T)$, $\delta_T = +0.124$. $\delta_T = +12.4\%$: $g_*(T_{\rm osc})$ (+5%), $T_{\rm osc}/\Lambda_{\rm QCD}$ (+4%), $m_a(T)$ (+3%).

Result. Numerical vs analytical gap: +12.4% (not +57%). Agreement with Planck at $\theta_i = 1.1162$ requires no tuning.

12 Test B.2: θ_i from inflation

$\delta\theta = H_{\rm inf}/(2\pi f_a)$, $\beta_{\rm iso} = (\delta\theta/\theta_i)^2$. Constraints: $r < 0.036 \Rightarrow H < 4.703 \times 10^{13}$ GeV; $\beta_{\rm iso} < 0.038 \Rightarrow H < 2.707 \times 10^{11}$ GeV.

Hypothesis $\theta_i \sim \delta\theta$ requires $H = 1.389 \times 10^{12}$ GeV, but $\beta_{\rm iso} = 1.0$ — **excluded at** 26σ. $\theta_i = 1.1162$ is a random draw from $[-\pi, \pi]$.

13 Test A.1: modular forms at $\tau = e^{i\pi/3}$

Details in §3.3, §3.4. Summary:

Table 8: Combinations of forms and deviation from $\alpha_G = 0.212207$.

Combination	Value	Deviation
$\lvert Y_1^{(6)}\rvert/\lvert Y_3^{(4)}\rvert$	0.212125	−0.0385%
$\lvert Y_2^{(6)}\rvert/\pi$	0.212411	+0.0965%
$\lvert Y_3^{(6)}\rvert/\pi$	0.216161	+1.86%
$\lvert Y_2^{(4)}\rvert/\pi$	0.221911	+4.57%

$|Y_2^{(6)}|^2 = 0.445302 \neq 4/9$; stable at $n = 5..30$. PSLQ finds nothing. p-value = 0.185 (not significant).

Conclusion. α_G remains a postulate.

14 Full balance

Table 9: Free parameters (13).

Category	Parameters	Count
Fermion moduli	τ_u, τ_d (complex)	4
up coefficients	a_u, b_u, c_u	3
down coefficients	a_1, a_2, b, c	4
Topological	$\mathcal{I}$	1
Axion	θ_i	1
Total		**13**

Table 10: Observables: fitted, predicted, and future work.

Sector	Observables	Status
Quark masses	$m_u/m_t, \ldots, m_b/m_t$	5 (fitted)
Quark CKM	$\lvert V_{us}\rvert, \lvert V_{cb}\rvert, \lvert V_{ub}\rvert$	3 (fitted)
Quark (J)	$\lvert J\rvert$	1 (fitted)
Axion	ν_a, $g_{a\gamma\gamma}$	predicted
Cosmology	$\Omega_a h^2$	1 (fitted via θ_i)
Lepton PMNS	$\theta_{12}, \theta_{13}, \theta_{23}, \lvert J\rvert$	4 (future work)
Fitted quarks		**9**

$$N_{\text{dof}}^{\text{quarks}} = 9 - 11 = -2. \tag{14.1}$$

14.1 Information criteria

Table 11: AIC/BIC ($n = 9$). The count $k = 13$ is the total number of model parameters; the quark-only fit uses $k_q = 11$ (see §14).

Model	k	χ^2	AIC	BIC
MEML 75	13	$\approx 10^{-20}$	26.0	28.6
SM	19	0.9	38.9	42.6

Formally, MEML wins. However $N_{\text{dof}} = -2$: $\chi^2 \sim 10^{-20}$ is a trivial consequence of over-determination, not of predictive power.

15 Open problems

1. **Coefficient degeneracy.** At fixed τ (2.3), different DE seeds find coefficients with $\chi^2 < 10^{-18}$, related by uniform scale $\times 1.87$. Check: $\chi^2(c) = \chi^2(kc) < 10^{-19}$ for any $k \in [0.5, 10]$ (Appendix D). Direct consequence of $N_{\text{dof}} = -2$. Priority for v8.0: mechanism to fix c_i (holomorphic 6D→4D couplings).
2. **Status of α_G.** Postulate. Systematic search yields no exact formula; near-coincidences are statistically expected ($p = 0.185$).
3. **Status of ε.** Ansatz; choice of 4π is not derived.
4. **Flatness of $\operatorname{Re}\tau$.** $\operatorname{Re}\tau_d$ is soft-flat (width 0.74); $\operatorname{Re}\tau_u$ is soft-flat (width 0.03). $\operatorname{Im}\tau$ is stable.
5. **Matching 6D↔4D.** Structural.
6. **LOO instability.** At $N_{\text{dof}} = -2$, spread of $\lvert V_{ub}\rvert$ from -97% to $+25407\%$.
7. **Over-determination in quarks.** 11 parameters for 9 observables.
8. **Universal ansatz excluded.** Only L6 yields ideal χ^2.
9. **GUT dependence on $\mathcal{I}$.** $M_6 \propto \mathcal{I}^{-1/4}$.
10. **Analytic vs numerical $\Omega_a h^2$.** $+12.4\%$, explained by T-effects.
11. **Derivation of θ_i.** Excluded at 26σ.

16 Roadmap

Table 12: Phases of development.

Phase	Task	Priority
I	Mechanism for c_i: holomorphic 6D→4D couplings	Critical
I	Flux stabilization of $\mathrm{Im}\,\tau$ ($\pm 3\%$)	High
II	Reduce $11 \to 9$ in quarks	Medium
II	Analytical derivation of c_i	Medium
III	ADMX, DESI, DUNE, KamLAND-Zen	External
IV	Leptogenesis, dark energy	Long-term

17 Conclusions

What has been done:

1. A model with 13 parameters for 15 observables.
2. 4 systematic tests (B.1, C.3, B.2, A.1) — all negative.
3. A global fit $\chi^2 = 6.195 \times 10^{-20}$ at $\tau_u = -0.1578 + 2.3852i$, $\tau_d = +0.3694 + 1.5410i$; independently reproduced (3 DE runs).
4. A test of coefficient degeneracy: χ^2 is invariant under $c_i \to k c_i$.
5. $\Omega_a h^2$: bare → anh → num, $\delta_T = +12.4\%$.
6. $g_{a\gamma\gamma}$ (KSVZ/DFSZ) — two-point target for ADMX.
7. AIC/BIC with $N_{\mathrm{dof}} < 0$ caveat.

What works: PMNS ($< 0.5\%$), CKM ($\chi^2 \sim 10^{-20}$, fit), Planck compatibility of dark matter, testable prediction $\nu_a = 6.8688$ GHz, $g_{a\gamma\gamma} \in \{+4.38 \times 10^{-15}, -1.13 \times 10^{-14}\}$ GeV^{-1}.

What does not work: LOO is unstable; coefficients are scale-degenerate; α_G, ε, θ_i are not derived from first principles.

Verdict. The model is *phenomenological*. It reproduces observations but does not predict. The coefficient degeneracy shows that the model interpolates 9 observations with 11 parameters.

Priority for future work: a mechanism to fix c_i. Flux is only needed for $\mathrm{Im}\,\tau$ at $\pm 3\%$.

A A_4 modular forms: full q-expansions

Definition: $q = e^{2\pi i \tau}$.

$$Y_1^{(2)} = 1 + \sum_{n=1}^{17} C_1^{(n)} q^n, \tag{A.1}$$

$$Y_2^{(2)} = -6\, q^{1/3} \left(1 + \sum_{n=1}^{11} C_2^{(n)} q^n\right), \tag{A.2}$$

$$Y_3^{(2)} = -18\, q^{2/3} \left(1 + \sum_{n=1}^{10} C_3^{(n)} q^n\right), \tag{A.3}$$

$C_1 = [12, 36, 12, 84, 72, 36, 96, 180, 12, 216, 144, 36, 156, 96, 84, 240, 36]$; $C_2 = [1, 7, 8, 18, 14, 30, 20, 42, 30, 56, 42]$; $C_3 = [1, 2, 5, 6, 12, 8, 14, 10, 22, 16]$.

Convolutions:

$$Y_1^{(4)} = (Y_1^{(2)})^2 - Y_2^{(2)} Y_3^{(2)}, \tag{A.4}$$

$$Y_2^{(4)} = (Y_3^{(2)})^2 - Y_1^{(2)} Y_2^{(2)}, \tag{A.5}$$

$$Y_3^{(4)} = (Y_2^{(2)})^2 - Y_1^{(2)} Y_3^{(2)}, \tag{A.6}$$

$$Y_1^{(6)} = \tfrac{1}{3}\left[2Y_1^{(2)} Y_1^{(4)} - Y_2^{(2)} Y_3^{(4)} - Y_3^{(2)} Y_2^{(4)}\right], \tag{A.7}$$

$$Y_2^{(6)} = \tfrac{1}{3}\left[2Y_3^{(2)} Y_3^{(4)} - Y_1^{(2)} Y_2^{(4)} - Y_2^{(2)} Y_1^{(4)}\right], \tag{A.8}$$

$$Y_3^{(6)} = \tfrac{1}{3}\left[2Y_2^{(2)} Y_2^{(4)} - Y_3^{(2)} Y_1^{(4)} - Y_1^{(2)} Y_3^{(4)}\right]. \tag{A.9}$$

B Numerical fit procedure

`differential_evolution`: 3 seeds (42, 43, 44); maxiter= 150; popsize= 15; tol= 10^{-8}; polish=False, then Nelder–Mead polish (maxiter= 10000).

Bounds: $\mathrm{Re}\,\tau \in [-0.5, 0.5]$, $\mathrm{Im}\,\tau_u \in [0.87, 2.5]$, $\mathrm{Im}\,\tau_d \in [0.87, 1.60]$, $c_i \in [10^{-8}, 10^2]$.

Observables: SVD $M_u = U_u^\dagger \Sigma_u V_u$, $M_d = U_d^\dagger \Sigma_d V_d$; masses $m_i = \Sigma_i$; CKM $V = U_u^\dagger U_d$; five mass ratios, $|V_{us}|$, $|V_{cb}|$, $|V_{ub}|$, $|J|$.

Runtime: 90–120 s per full run (Python 3.10 + scipy).

C Reproduction of τ and χ^2

Table 13: Three independent DE runs.

Source	$\mathrm{Re}\,\tau_u$	$\mathrm{Im}\,\tau_u$	$\mathrm{Re}\,\tau_d$	$\mathrm{Im}\,\tau_d$
Run 1	−0.15779	2.38518	+0.36939	1.54105
Run 2	−0.15779	2.38518	+0.36939	1.54105
Run 3	−0.12893	2.38411	−0.37212	1.54102

$\mathrm{Im}\,\tau_u = 2.385 \pm 0.001$; $\mathrm{Im}\,\tau_d = 1.541 \pm 0.0001$.

Rounding sensitivity:

Table 14: Sensitivity of χ^2 to τ rounding.

Precision of τ	χ^2
exact (10 digits)	6.195×10^{-20}
4 significant digits	1.08×10^{-3}
3 significant digits	2.55×10^{-3}

D Proof of coefficient degeneracy

At fixed τ (2.3), multiply all c_i by k:

Table 15: Invariance of χ^2 under $c_i \to kc_i$.

k	χ^2
0.5	6.20×10^{-20}
1.0	6.20×10^{-20}
1.87	1.08×10^{-19}
2.0	6.20×10^{-20}
5.0	3.00×10^{-19}
10.0	3.00×10^{-19}

All $\chi^2 < 10^{-18}$: $\chi^2(kc) \approx \chi^2(c)$. **Uniform scale degeneracy proven.**

E Flatness of $\mathrm{Re}\,\tau$

Scan in $\mathrm{Re}\,\tau_u$ (at $\mathrm{Re}\,\tau_d = +0.369$, coefficients fixed): width of $\chi^2 < 10$: $[-0.40, +0.00]$ (0.40).

Scan in $\mathrm{Re}\,\tau_d$**:** width of $\chi^2 < 10$: $[+0.30, +0.50]$ (0.20).

At fixed coefficients $\mathrm{Re}\,\tau$ is soft-flat. At refit (Appendix C) $\mathrm{Re}\,\tau_d \in [-0.37, +0.37]$.

F LOO test: procedure and results

For each $j \in \{1, \ldots, 9\}$: remove the j-th observable, fit $8 \to 7$ coefficients, predict the j-th. $N_{\mathrm{dof}} = +1$.

Results U6×D6: mean $|V_{ub}|$ +16728%, std 11819%; $|J|$ +3598%, std 7487%.

Direct relation to degeneracy (Appendix D).

G Systematic search for α_G

Basis: 16 elements ($Y_i^{(2)}$, $Y_i^{(4)}$, $Y_i^{(6)}$, η, π, $\sqrt{\pi}$, π^2, $e^{-\pi^2}$), where $\eta(\tau) = q^{1/24}\prod_{n=1}^{\infty}(1-q^n)$ is the Dedekind eta function. Tested $N = 392$ combinations: $|a|/|b|$ (240), $|a| \cdot |b|$ (120), $\sqrt{|a|/\pi}$ (16), $|a|^2/\pi^{3/2}$ (16).

2 coincidences below 0.1%**:** $|Y_1^{(6)}|/|Y_3^{(4)}| = 0.212125$ (-0.0385%, $n = 5..30$); $|Y_2^{(6)}|/\pi = 0.212411$ ($+0.0965\%$, $n = 5..30$).

Poisson: $\langle n \rangle = 392 \cdot 0.002 = 0.78$; $n_{\mathrm{obs}} = 2$; $P(n \geq 2) = 0.185$ (not significant).

Conclusion: α_G remains a postulate.

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